

FPGA-Based Traffic Light Controller with Priority System

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Abstract

Traffic congestion and emergency vehicle delays are common problems in urban transportation systems. This project presents an FPGA-Based Traffic Light Controller with an Emergency Vehicle Priority System using Verilog HDL. The controller manages Red, Yellow, and Green traffic signals through a Finite State Machine (FSM). When an emergency vehicle is detected, the controller overrides the normal sequence and immediately provides a Green signal to ensure safe and rapid passage.

Introduction

Traffic signal controllers play a vital role in road safety and transportation efficiency. Traditional systems operate on fixed timing schedules and cannot dynamically respond to emergency situations. This project introduces a digital logic-based traffic light controller capable of prioritizing emergency vehicles while maintaining standard traffic operations.

Objectives

- Design a traffic light controller using Verilog HDL.
 - Implement FSM-based signal transitions.
 - Provide emergency vehicle priority override.
 - Develop a simulation-ready FPGA-compatible design.
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System Architecture

Inputs

- Clock Signal (clk)
- Reset Signal (reset)
- Emergency Vehicle Signal (emergency)

Outputs

- Red Light
 - Yellow Light
 - Green Light
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Working Principle

Normal Mode

The controller cycles through: 1. Red State 2. Green State 3. Yellow State

Emergency Mode

When the emergency signal becomes active:

- Normal traffic sequence is interrupted.
 - Green light is activated immediately.
 - Emergency vehicle receives priority passage.
 - Normal sequence resumes after emergency clearance.
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Implementation

The system is implemented using Verilog HDL. A Finite State Machine controls state transitions. Timers determine the duration of each traffic light state. Emergency detection logic overrides the FSM output and forces the Green signal.

Advantages

- Fast emergency response support.
 - Improved traffic management.
 - FPGA implementation capability.
 - Simple and scalable architecture.
 - Enhanced road safety.
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Applications

- Smart Traffic Systems
 - Smart Cities
 - Highway Traffic Control
 - Emergency Vehicle Management
 - Intelligent Transportation Systems
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Results

The Verilog implementation successfully demonstrated traffic light sequencing and emergency priority functionality. The system correctly transitioned between Red, Green, and Yellow states and immediately switched to Green during emergency conditions.

Future Scope

- IoT-enabled traffic monitoring.
 - Vehicle density-based signal control.
 - AI-assisted traffic optimization.
 - Cloud-connected smart transportation systems.
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Conclusion

The FPGA-Based Traffic Light Controller with Priority System was successfully designed and implemented using Verilog HDL. The controller effectively manages traffic flow while providing emergency vehicle priority, making it suitable for modern intelligent transportation applications.

References

1. Verilog HDL Fundamentals.
2. Digital Logic Design Principles.
3. FPGA-Based System Design Concepts.
4. Intelligent Transportation System Research Papers.